Computational Supplement to "A One-Loop BV Obstruction in Pure Weyl Gravity and Its Formal τ-adic Wess–Zumino Resolution"

Asger Alstrup Palm

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Abstract

This supplement records the exact finite computations underlying the strict one-loop BV obstruction and its formal τ-adic compensator resolution. It separates basis exhaustiveness, relative-cohomology ranks, antifield and gauge-fixing transport, analytic coefficient provenance, the compensator cotangent lift, quartet contraction, the extended quotient, and QME restoration. Tables and matrices carrying numerical or discrete claims are generated directly from content-addressed JSON certificates. Independent verifiers reject stale generated material, altered ranks, missing classes, over-promoted lifecycle states, and dependency-hash drift.

Scope and dependency labels

The supplement proves only the claims labeled LOCAL-ALGEBRAIC and EUCLIDEAN-SPECTRAL in the companion manuscript. It does not prove a Lorentzian QME, renormalized causal products, a Hadamard state, positivity, particles, residual transfer, or an all-loop theorem. The strict and compensator-extended theories remain different rows throughout.

Contents

1 Certificate graph

The calculation is organized as a directed acyclic graph rather than a single symbolic notebook:

$$\begin{array}{c} \text{canonical AFN0 bases and descents} \longrightarrow \text{minimal Koszul–Tate collapse} \\ \downarrow \\ \text{nonminimal/gauge-fixed quotient plus analytic coefficients} \\ \downarrow \\ \text{regulated Slavnov class} \longrightarrow \text{strict QME disposition,} \end{array}$$

and independently

$$\begin{array}{c} \text{compensator master term} \longrightarrow \text{cotangent lift and quartet} \\ \downarrow \\ \text{dressed pure-Diff quotient} \longrightarrow \text{extended QME disposition.} \end{array}$$

No determinant row is promoted directly into BV cohomology. The analytic coefficient vector is accepted only after it is reduced against the complete gauge-fixed quotient with the same normalization and classical snapshot.

2 Exact strict quotient

2.1 Declared local algebra

The production slice is four-dimensional, form degree four, engineering dimension four, and ghost number zero or one. Tensor generators include the metric, curvature and covariant derivatives, the Diff and Weyl ghosts, the minimal antifields, and the declared nonminimal doublets. Epsilon tensors are included in the odd sector. The canonical quotient imposes:

- algebraic and differential Bianchi identities;
- tensor symmetries and four-dimensional antisymmetrization;
- graded factor permutations and Grassmann signs;
- dummy-index graph isomorphism;
- integration by parts and horizontal exactness.

The basis generator is checked twice: forward tensor-graph generation and reverse coverage of every grading-admissible signature. A class receives a complete nontriviality witness only after the top, lower-form cocycle, and lower-form boundary spaces are exhaustive.

2.2 Orbit-first reduction

The even ghost-one basis has seven canonical top carriers before closure and boundary reduction. Orbit-first contraction resolves the two formerly pending mixed signatures by materializing only thirty raw pairings. The odd calculation resolves all three pending mixed signatures: forty-five raw graphs reduce to nine signed orbits, all of which vanish by the Bianchi relations. The single odd Weyl pseudoscalar remains.

The exact rank data and final dimensions are generated below.

Stage	$\dim H^{0,4}_{\mathrm{even}}$	$\dim H^{0,4}_{\mathrm{odd}}$	$\dim H^{1,4}_{\mathrm{even}}$	$\dim H^{1,4}_{\mathrm{odd}}$
strict gauge-fixed BV	2	1	2	1
τ-adic extended BV	3	1	0	0

Table 1: Exact dimension-four local BV quotient dimensions.

$$Q_W = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \qquad h = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \qquad Q_W h + h Q_W = I_4, \tag{1}$$

in the ordered basis $(\tau, \omega, \omega^*, \widehat{\tau}^*)$. The certificate checks 24 component rows on 8 generators and their jet prolongations over Q; all component gradings, δ^2, $\delta\gamma + \gamma\delta$, and Q^2 pass exactly.

The Euler primitive used by the boundary matrix is

$$B_E = \int \sqrt{g}\, \left[\tau E_4 + 4G^{\mu\nu}\nabla_\mu\tau\nabla_\nu\tau - 4(\Box\tau)(\nabla\tau)^2 + 2(\nabla\tau)^4\right], \qquad Q_W B_E = \int \sqrt{g}\, \omega E_4 \pmod{\mathrm{d}_h}, \tag{2}$$

with the sign convention $Q_W B_C = \int \sqrt{g}\, \omega C^2$ and $Q_W B_E = \int \sqrt{g}\, \omega E_4$ modulo d_h.

Spectral-sequence step	Exact input	Rank/count	Outcome
Koszul–Tate E_0	adapted regular-Bach pairs	6 pairs	positive AFN columns vanish
$E_1 \Rightarrow E_2$	AFN0 modulo Euler–Noether ideal	9016 monomials	collapse at E_2
pure-Diff/mixed total complex	refined ambient signatures	720 signatures	zero additional quotient
nonminimal extension	pointwise Diff×Weyl doublets	10 pairs	chain contraction
gauge-fixing transport	local BV-canonical similarity	25080 monomials	quotient preserved

Table 2: Strict-quotient spectral-sequence and contraction ledger on the regular Bach locus.

Sector	raw graphs/pairings materialized	closure rank	boundary rank
even AFN0 relative carrier	30	6	4
odd AFN0 mixed carrier	45	1	0

Table 3: Orbit-first strict AFN0 quotient audit. The ambient 2.86×10^9 raw-graph expansion is never materialized.

$$B_{\text{ext}} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \qquad \operatorname{rank} B_{\text{ext}} = 4, \tag{3}$$

from the ordered primitive basis $(B_C, B_E, B_P, B_\square)$ to $(\omega C^2, \omega E_4, \omega C\widetilde{C}, \omega\square R)$. The strict breaking and its extended boundary image are both

$$\left(\frac{199}{30}, -\frac{87}{20}, 0, 0\right).$$

2.3 Antifields and gauge fixing

The imported minimal classical export supplies every Koszul–Tate row. On the regular Bach locus the positive-antifield sector is an explicit sum of adapted pairs, and its contracting homotopy verifies

$$\delta\sigma + \sigma\delta = N_{\text{pair}}.$$

Consequently the AFN0 Weyl representatives lift through the minimal spectral sequence. A separate total-complex computation proves that the pure-Diff and mixed Diff–Weyl sectors add no independent dimension-four class.

Concretely, the six adapted pairs are (g^*, E_g), (ξ^*, N_ξ), (ω^*, N_ω) and their Lie-prolonged partners. The E_0 positive-antifield columns vanish, E_1 is AFN0 modulo the Euler–Noether ideal, and the sequence collapses at E_2. Since every positive-antifield generator in this slice has form degree four, no omitted lower-form positive-AFN carrier can alter the relative quotient. This is a chart theorem: replacing highest TT metric jets by the Bach Euler row assumes the declared regular Bach locus.

Ten general nonminimal pairs then contract on 25,080 graded regression monomials. Their covariant jet prolongations commute with d_h, and a free-word normal-form computation transports

Sector	operator	rank	statistics	exponent in Z	zero modes removed	local prescription
metric TT, upper	`Delta_2_perp(4)`	5	bosonic	$-\frac{1}{2}$	0	factorwise spectral cut
Diff–Weyl scalar ghost	`Delta_0(-4)`	1	fermionic	$\frac{1}{2}$	5	cut; negative-mode phase locally constant
metric TT, lower	`Delta_2_perp(2)`	5	bosonic	$-\frac{1}{2}$	0	factorwise spectral cut
transverse Diff ghost	`Delta_1_perp(-3)`	3	fermionic	$\frac{1}{2}$	10	factorwise spectral cut

Table 4: Human-readable determinant ledger for the accepted repository Euclidean integration slice. Exponents are determinant powers in Z; the corresponding Γ_1 powers have the opposite sign.

Sector	C^2	E_4	$C\widetilde{C}$	$\Box R$
metric TT, upper	$\frac{37}{24}$	$-\frac{31}{72}$	0	0
Diff–Weyl scalar ghost	$\frac{89}{120}$	$-\frac{269}{360}$	0	0
metric TT, lower	$\frac{29}{8}$	$-\frac{181}{72}$	0	0
transverse Diff ghost	$\frac{29}{40}$	$-\frac{79}{120}$	0	0
exact sum	$\frac{199}{30}$	$-\frac{87}{20}$	0	0

Table 5: Factorwise local heat-kernel coordinates after the accepted determinant exponents, measure and quartet cancellations. Their exact sum is the bosonic-Weyl-parameter Ward carrier before ghost replacement and BV quotient reduction.

the SDR through an arbitrary invertible local BV-canonical gauge-fixing transformation. Thus the minimal, nonminimal, and gauge-fixed quotient dimensions agree. This explicit chain comparison is why the main theorem is not labeled AFN0-only.

3 Coefficient and Slavnov route

3.1 Analytic input

The accepted Euclidean principal-symbol sequence is

$$(T \oplus \mathbf{1})_5 \xrightarrow{\sigma(K)} (S^2T^*)_{10} \xrightarrow{\sigma(P_{\mathrm{TT}})} (S^2_{\mathrm{TT}}T^*)_5.$$

The first bundle contains four Diff plus one Weyl gauge parameter; the middle bundle is the metric perturbation; the last is the TT symbol slice, not a five-component claim about the unreduced Bach equation. It has ranks $(5,5)$ at nonzero covector, zero composition, and exact image–kernel equality. The formal adjoint sequence, Diff/Weyl nonminimal doublets, and four kinetic blocks replay independently. The determinant bundle ranks and exponents are

$$(5,1,5,3), \qquad \left(-\frac{1}{2},+\frac{1}{2},-\frac{1}{2},+\frac{1}{2}\right),$$

Density coordinate	repository one-loop coefficient
C^2	$\frac{199}{30}$
E_4	$-\frac{87}{20}$
$C\widetilde{C}$	0
$\Box R$	0

Table 6: Coefficient vector in the convention $(4\pi)^{-2}[cC^2 - aE_4 + pC\widetilde{C} + b\Box R]$.

Standard-sign field	C^2 coordinate	E_4 coordinate
pure Weyl graviton	$\frac{199}{30}$	$-\frac{87}{20}$
real conformal scalar	$\frac{1}{120}$	$-\frac{1}{360}$
Weyl fermion	$\frac{1}{40}$	$-\frac{11}{720}$
Dirac fermion	$\frac{1}{20}$	$-\frac{11}{360}$
gauge vector (with BRST ghosts)	$\frac{1}{10}$	$-\frac{31}{180}$

Table 7: Exact anomaly vectors in the convention $(c, -a)$. The functional $(1, 0)$ is strictly positive on the gravity vector and every allowed matter ray, already excluding cancellation in the nonnegative cone.

with priming dimensions $(0, 5, 0, 10)$. The generated determinant table prints each operator, differential order, rank, statistics, exponent, and zero-mode subtraction separately. Every displayed factor is second order; the fourth-order TT determinant is the product of its two displayed second-order factors.

For each factor, the two invariant coordinates are reconstructed from independent carriers: the Ricci-flat row supplies $c_i - a_i$ and the round-S^4 row supplies a_i, so the generated table forms $c_i = (c_i - a_i) + a_i$ before summing factors. The parity-odd zero is factorwise for the declared real parity-even operator family. The $\Box R = 0$ column is instead the chosen scheme for the total functional, implemented by one local R^2 counterterm; no factorwise scheme-invariant type-D claim is made.

The local C^2 coordinate is exposed on a non-conformally-flat Ricci-flat carrier. The Euler coordinate is independently fixed on round S^4. The measure ledger includes York and ghost super-Jacobians, all nonminimal quartets, conformal-Killing zero modes, the oriented auxiliary contour, and the local parity Ward identity. Global determinant phase and conformal-group volume are not used to infer a local coefficient.

3.2 Cohomological reduction

Let the ordered strict basis be

$$(\omega C^2, \omega E_4, \omega C\widetilde{C}, \omega\Box R).$$

The raw analytic vector is the one displayed in Table 6. The fourth coordinate is removed with the stored local identity

$$\sqrt{g}\,\omega\Box R\,\mathrm{d}^4x = -\frac{1}{12}s\big(\sqrt{g}\,R^2\,\mathrm{d}^4x\big) - \mathrm{d}\star\big(R\,\mathrm{d}\omega - \omega\,\mathrm{d}R\big).$$

Anomaly-induced functional carrier	exact coefficient
$\langle \mathcal{E}_4, G_4 C^2 \rangle$	$\frac{199}{120}$
$\langle \mathcal{E}_4, G_4 \mathcal{E}_4 \rangle$	$-\frac{87}{160}$
$\int \sqrt{g} R^2$	$\frac{29}{120}$

Table 8: Exact Paneitz/Riegert solve. The Weyl-response matrix is $\mathrm{diag}(4, 8, -12)$; re-expansion returns $(199/30, -87/20, 0)$ in $(C^2, E_4, \Box R)$.

Flat-TT form-factor datum	exact value
c	$\frac{199}{30}$
β_2	$\frac{199}{15}$
$A_{\log}$	$-\frac{199}{60}$
$\mu\partial_\mu F_C$	$\frac{199}{30}$
finite local C^2 constant	open

Table 9: Universal nonzero-momentum flat Euclidean TT logarithmic form factor. The additive finite normalization and curved completion are not fixed.

The first two normalized dual functionals evaluate to $199/30$ and $-87/20$, while the odd functional evaluates to zero by the Ward identity. Therefore

$$[\mathcal{A}^{(1)}] = \frac{1}{(4\pi)^2}\left(\frac{199}{30}[\omega C^2] - \frac{87}{20}[\omega E_4]\right) \neq 0.$$

The strict QME verdict is determined by this nonzero cohomology class. The beta function, determinant logarithm, and background trace anomaly are not used as substitutes for the regulated Slavnov insertion.

The bridge is the regulated identity

$$\mathcal{S}_S \Gamma^{(1)}_{\mathrm{reg}} = \lim_{d\to 4} s_d \left[\frac{\mu^{d-4}}{(4\pi)^2(d-4)} \int \sqrt{g}\,(cC^2 - aE_4 + b\Box R + \text{exact rows})\right].$$

The $(d-4)$ Weyl variation of the continued densities cancels the pole. Replacing the bosonic Weyl parameter by ω, adjoining the certified universal Diff descent, and reducing exact, parameter-dependent, and total derivative rows produces the ghost-one insertion. Thus the determinant ledger supplies coefficients to an independently assembled Slavnov carrier; it is not itself called a BV anomaly.

In certificate notation the residue map is

$$s_d I_i(d) = (d-4)A_i^4 + s_d B_i(d) + \mathrm{d}_h C_i(d) + O((d-4)^2),$$

with A_i^4 the complete ghost-one carrier. The pole residue is paired with the normalized quotient duals only after the exact and horizontal rows are removed. This is the machine-readable distinction among determinant coefficients, a bosonic Weyl Ward variation, and the BV Slavnov class.

3.3 Matter cone

For standard-sign free conformal scalars, Weyl/Dirac fermions, and gauge vectors, cancellation is an exact rational cone-feasibility problem. A separating dual covector is positive on every allowed matter

Covariant-log datum	exact value
curvature order of $\langle C, \log(\Delta_C/\mu^2)C\rangle$	2
first admissible operator-choice difference	3
logarithmic coefficient	$-\frac{199}{60}$
selected FV Weyl-orbit completion	certified
first forced FV correction order	3
independent cubic Weyl-invariant form factors	open
finite C^2/R^2 normalization	open

Table 10: Covariant curvature-squared logarithm, its exact FV completion as a selected carrier, and the independent-data boundary. The coefficient remains $-\frac{199}{60}$.

ray and on the pure-gravity breaking. Hence no nonnegative real multiplicity vector, and therefore no nonnegative integral multiplicity vector, cancels both even coordinates. This theorem deliberately excludes nonunitary signs, interacting fixed points, conformal higher spin, and compensators.

4 Compensator BV extension

4.1 Rows derived from one master term

The extension is generated from

$$S_{\tau,\min} = \int \tau^*(\xi^\mu \nabla_\mu \tau + \omega).$$

Its four Euler derivatives force

$$\begin{aligned}
Q\tau &= \mathcal{L}_\xi \tau + \omega, \\
\delta\omega^* &= N_\omega + \tau^*, \\
\delta\xi^*_\mu &= N_{\xi,\mu} + \tau^* \nabla_\mu \tau, \\
\gamma\tau^* &= \mathcal{L}_\xi \tau^*.
\end{aligned}$$

The Lie-prolonged cotangent rows carry the opposite formal-adjoint signs. No coefficient is fitted to a nilpotency defect.

On all twenty-four strict-plus-compensator generators and jet generators the independent replay finds $\delta^2 = 0$, $\delta\gamma + \gamma\delta = 0$, and $Q^2 = 0$. It also checks every ghost number, antifield number, parity, and tensor component type.

4.2 Dressed canonical coordinates

The change

$$\widehat{g} = e^{-2\tau} g, \qquad \widehat{g}^* = e^{2\tau} g^*, qquad \widehat{\tau}^* = \tau^* + 2g_{\mu\nu} g^{*\mu\nu}$$

obeys the canonical one-form identity

$$g^*\delta g + \tau^*\delta\tau = \widehat{g}^*\delta\widehat{g} + \widehat{\tau}^*\delta\tau.$$

It sends the Weyl Noether row to $\delta\omega^* = \widehat{\tau}^*$ and yields the quartet matrices in Equation 1.

Certificate	SHA-256 prefix
quantum-weyl/local_bv/certificates/AFNO_H14_EVEN_CANONICAL_QUOTIENT.json	6bf8b7ef099ce40b
quantum-weyl/local_bv/certificates/AFNO_H14_ODD_CANONICAL_QUOTIENT.json	0134e28cf3fd0bf4
quantum-weyl/local_bv/certificates/AFNO_DIFF_MIXED_MINIMAL_BV_H14.json	f2624bad4840e336
quantum-weyl/local_bv/certificates/GENERAL_NONMINIMAL_GAUGE_FIXED_CONTRACTION.json	4513be4824760577
quantum-weyl/local_bv/certificates/MINIMAL_BV_KOSZUL_TATE_COLLAPSE.json	f2b04706724d79b5
quantum-weyl/spectral/euclidean/certificates/REPOSITORY_EUCLIDEAN_ELLIPTIC_COMPLEX.json	a1eb7dd598ed3417
quantum-weyl/spectral/euclidean/certificates/REPOSITORY_FULL_BV_MULTIPLICITY_LEDGER.json	5bdc6ee4674e3a58
quantum-weyl/spectral/euclidean/certificates/STANDARD_EUCLIDEAN_LOCAL_B4_INTEGRATION_SLICE.json	edde6206a9dc62d0
quantum-weyl/spectral/euclidean/certificates/REPOSITORY_NONCONFORMALLY_FLAT_OR_RICCI_FLAT_FULL_BV_OPERATOR_MEASURE_COEFFICIENT_MATCH.json	d23543cdab828288
quantum-weyl/anomalies/certificates/REGULATED_REPOSITORY_BV_SLAVNOV_BREAKING.json	d11ebc782c6a388e
quantum-weyl/anomalies/certificates/UNITARY_CONFORMAL_MATTER_CANCELLATION_NO_GO.json	9c4b1218450e20f0
quantum-weyl/anomalies/certificates/WESS_ZUMINO_MINIMAL_BV_COTANGENT_LIFT.json	b265dc9d86938ed7
quantum-weyl/anomalies/certificates/WESS_ZUMINO_COMPENSATOR_EXTENSION_PREFLIGHT.json	10dbd0b0a1e9190d
quantum-weyl/anomalies/certificates/WESS_ZUMINO_EXTENDED_LOCAL_BV_COHOMOLOGY.json	fa21fc6071ae5227
quantum-weyl/transfer/certificates/ANOMALY_INDUCED_NONLOCAL_GAMMA1.json	9531ef49d0d930fc
quantum-weyl/transfer/certificates/FLAT_TT_LOGARITHMIC_GAMMA1.json	8d4bc8bdf702cf3c
quantum-weyl/transfer/certificates/CURVATURE_SQUARED_COVARIANT_LOG_GAMMA1.json	a67a475da6352ad1
quantum-weyl/transfer/certificates/FV_CONFORMIZED_C2_LOG_GAMMA1.json	7a4e55efa7284d30

Table 11: Content-addressed inputs to the generated supplement tables. Full hashes are stored in the Paper 12 claim map.

Equivalently, $h\omega = \tau$, $h\widehat{\tau}^* = \omega^*$ and h vanishes on τ, ω^*. Extending h to covariant jets as a graded derivation and dividing by positive quartet number gives the contraction $Q_W h + hQ_W = N_{\text{quartet}}$. Diff transport and d_h preserve the filtration, so homological perturbation lifts this contraction to the total formal complex.

More directly, write the total differential as $\mathcal{D} = Q_W + \delta$. In dressed covariant jets the quartet members occur in matched Diff modules, so the covariant graded derivation h obeys $[\delta, h]_+ = 0$ and $[N_{\text{quartet}}, \delta] = 0$. Thus

$$\mathcal{D}h + h\mathcal{D} = N_{\text{quartet}}.$$

Every closed component of positive quartet number n is explicitly the boundary $\mathcal{D}(n^{-1}h\mathbf{a}_n)$. At quartet number zero the total differential is the pure-Diff BV differential of $\widehat{g}$. The independently certified pure-Diff ghost-one zero, rather than the four primitive matrix alone, therefore completes the proof of $H^{1,4}_{\text{ext}} = 0$.

The coordinate change is interpreted in the formal τ-adic local analytic completion. Exact inverse series are checked through order eight, and the all-orders coefficient convolution is

$$\sum_{k=0}^{n} \frac{(-2)^k}{k!} \frac{2^{n-k}}{(n-k)!} = \frac{2^n}{n!}(1-1)^n = \delta_{n0}.$$

5 Extended quotient and restoration

Quartet contraction reduces the extended complex to pure Diff BV cohomology in $\widehat{g}$. The dimension-four ghost-zero basis therefore contains the new invariant $R(\widehat{g})^2$ in addition to $C(\widehat{g})^2$, E_4, and the odd Pontryagin density. $\Box R(\widehat{g})$ remains horizontally exact.

The only dimension-zero natural symmetric tensor that could generate an incoming positive-antifield boundary is $\widehat{g}$ itself. Its image is the trace of the Bach Euler row and vanishes. Normalized coordinate duals for all four surviving ghost-zero classes annihilate both the horizontal and positive-antifield boundary spaces.

At ghost number one the primitives $(B_C, B_E, B_P, B_\Box)$ map to the complete strict four-candidate carrier by the identity matrix in Equation 3. The independently certified pure-Diff anomaly quotient is zero. Hence the full extended dimension-four $H^{1,4}$ vanishes in both parities.

The coefficient-bearing counterterm is

$$S_{\mathrm{ct}}^{(1)} = -\frac{\hbar}{(4\pi)^2}\left(\frac{199}{30}B_C - \frac{87}{20}B_E\right),$$

and exact coordinate replay gives

$$\mathcal{A}^{(1)} + s_{\mathrm{ext}} S_{\mathrm{ct}}^{(1)} = 0 \pmod{\mathrm{d}_h}.$$

This is the one-loop local Euclidean QME restoration theorem in the declared formal τ-adic theory. It changes the theory: setting $\tau = 0$ does not make the strict quotient anomaly-free, and the dressed algebra contains the new $R(\widehat{g})^2$ counterterm direction.

6 First Slavnov operator: fixed part and exact ambiguity

The counterterm fixes the local Hamiltonian contribution

$$Q_1^{\mathrm{WZ}} = (C_1^{\mathrm{WZ}}, \cdot).$$

Here

$$C_1^{\mathrm{WZ}} = -(4\pi)^{-2}\left(\frac{199}{30}B_C - \frac{87}{20}B_E\right).$$

Since C_1^{WZ} has antifield number zero, its field and ghost rows vanish and its cotangent rows are the Euler derivatives with respect to $\widehat{g}$ and τ. This is an actual coefficient-bearing part of Q_1, but not the complete renormalized operator.

The exact non-uniqueness witness uses flat Euclidean momentum $p = (1,0,0,0)$, a TT polarization $h_{11} = 1, h_{22} = -1$, and the conformal polarization $h_{\mu\nu} = \delta_{\mu\nu}$. Direct contraction of the linearized Riemann tensor gives

	$C^{(1)2}$	$R^{(1)2}$
TT	1	0
conformal	0	9

Including the topological Euler and Pontryagin columns, the allowed extended $H^{0,4}$ response matrix is

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 9 & 0 \end{pmatrix}, \qquad \text{rank} = 2.$$

The topological columns have zero compactly supported bulk Euler derivative; their boundary and global content is retained separately. Therefore finite $C(\widehat{g})^2$ and $R(\widehat{g})^2$ counterterms change the bulk Hamiltonian vector field independently. This receipt alone does not supply a nonlocal Γ_1 representative, the renormalized BV Laplacian or time-ordered product, or finite normalization conditions. The follow-up below supplies the anomaly-induced component but not the Weyl-invariant finite remainder. Complete Q_1 remains fail-closed as `NO_CERTIFIED_OPERATOR`.

6.1 Anomaly-induced nonlocal representative

The follow-up receipt solves the exact Weyl-response system rather than declaring the whole finite effective action absent. With

$$\mathcal{E}_4 = E_4 - \frac{2}{3}\Box R, \qquad \Delta_4 = \Box^2 + 2R^{\mu\nu}\nabla_\mu\nabla_\nu - \frac{2}{3}R\Box + \frac{1}{3}(\nabla^\mu R)\nabla_\mu,$$

the carrier basis

$$\left(\langle \mathcal{E}_4, G_4 C^2\rangle, \langle \mathcal{E}_4, G_4 \mathcal{E}_4\rangle, \int \sqrt{g} R^2\right)$$

has Weyl-response matrix $\mathrm{diag}(4, 8, -12)$. Exact rational elimination against the modified-basis target $(199/30, -87/20, -29/10)$ gives

$$\left(\frac{199}{120}, -\frac{87}{160}, \frac{29}{120}\right).$$

Re-expansion in $(C^2, E_4, \Box R)$ returns $(199/30, -87/20, 0)$ exactly. Thus

$$\Gamma_{1,\text{anom}} = \frac{1}{(4\pi)^2}\left\{\frac{1}{8}\langle \mathcal{E}_4, G_4(2cC^2 - a\mathcal{E}_4)\rangle + \frac{a}{18}\int \sqrt{g} R^2\right\}$$

is one anomaly-induced Euclidean representative.

The inverse contract is fail-closed. Either the Euclidean boundary problem is invertible, or G_4 is a self-adjoint generalized inverse and every displayed source lies in the complement of $\ker \Delta_4$. A compact Paneitz zero mode is not silently discarded: its global anomaly, homogeneous solutions, and boundary transgressions remain separate data. The Weyl-invariant finite functional, finite C^2 and absolute dressed $R(\widehat{g})^2$ normalization, and renormalized BV product are still absent, so this result does not promote complete Γ_1 or Q_1.

6.2 Universal nonzero-momentum flat-TT logarithm

The conformal-sector receipt next fixes the part left invariant by the inhomogeneous anomaly solve at quadratic order on a declared flat Euclidean TT carrier. With the fixture normalized by $\langle C^{(1)}, C^{(1)}\rangle = 1$, exact rational arithmetic gives

$$F_C(p^2;\mu) = -\frac{199}{60}\log\frac{p^2}{\mu^2} + z_C(\mu), \qquad -\frac{199}{60} = -\frac{c}{2} = -\frac{\beta_2}{4}.$$

The independent checks replay

$$\mu\partial_\mu F_C = \frac{199}{30}, \qquad F_C(p^2) - F_C(q^2) = -\frac{199}{60}\log\frac{p^2}{q^2},$$

and verify that the prior Riegert representative begins at $O(h^4)$ on this pure-TT slice. A finite reference condition only determines the family

$$z_C(\mu) = N_C + \frac{199}{60}\log\frac{\kappa_{\mathrm{ref}}^2}{\mu^2};$$

neither N_C nor κ_{ref} is selected by the anomaly. The receipt therefore rejects promotion to a curved completion, a zero-momentum result, a Lorentzian branch prescription, complete Γ_1/Q_1, or a particle entry.

6.3 Curvature-squared covariantization

The next receipt replaces the flat momentum logarithm by the spectral logarithm of a positive self-adjoint Laplace-type operator Δ_C on the Weyl-tensor bundle. On the complement of its kernel,

$$\Gamma^{(2)}_{1,C^2\log} = -\frac{1}{(4\pi)^2}\frac{199}{60}\langle C, \Pi_\perp \log(\Delta_C/\mu^2)\Pi_\perp C\rangle.$$

The verifier replays the coefficient and scale response and applies the curvature filtration to the resolvent Fréchet derivative. For $\Delta_C' = \Delta_C + V_1 + O(\mathcal{R}^2)$,

$$d\log(\Delta_C)[V_1] = \int_0^\infty (\Delta_C + s)^{-1} V_1 (\Delta_C + s)^{-1}\, ds,$$

so the sandwiched change has exact order $1+1+1=3$. Operator-choice dependence for this C^2 carrier therefore starts with cubic form factors within the class sharing principal symbol, connection, domain, and source complement and differing at first curvature order only by the self-adjoint endomorphism V_1. Connection variations, which can introduce first-order differential terms, are not covered. Kernel and boundary data, derivative-decorated nonlocal cubic form factors, finite C^2 constant and absolute dressed $R(\widehat{g})^2$ normalization remain fail-closed. The selected carrier's Weyl-orbit completion is handled separately below.

6.4 Fradkin–Vilkovisky completion of the selected carrier

The conformization receipt constructs

$$L_g = \Box_g - \frac{1}{6}R_g, \qquad u_g = 1 + \frac{1}{6}L_g^{-1}R_g, \qquad \bar{g}[g] = u_g^2 g.$$

It independently checks $L_g u_g = 0$, $R[\bar{g}] = 0$, the Weyl-weight ledger $2(-1)+2=0$, and the boundary-normalized transformation $u_{e^{2\sigma}g} = e^{-\sigma}u_g$. Consequently $\bar{g}[e^{2\sigma}g] = \bar{g}[g]$, and evaluation of the full selected $C\log\Delta_C C$ carrier—its tensors, pairing, projector, domain and logarithm—on $\bar{g}$ gives an exact nonlocal functional invariant under the declared local Weyl transformations.

The verifier retains the inverse-domain hypothesis, replays the coefficient $-199/60$ and the first correction order three, and rejects promotion to derivative-decorated nonlocal cubic form factors, complete Γ_1/Q_1, residual transfer or a Lorentzian result. It also enforces a carrier crosswalk: the nonlocal $\bar{g}[g]$ is distinct from the local τ-adic field $\widehat{g}[g,\tau]$.

6.5 FV anomaly action and dependent Ricci sector

The coefficient-bearing anomaly action is replayed as

$$\Gamma_{1,\mathrm{FV}} = \frac{1}{(4\pi)^2}\int\sqrt{g}\left[\left(\frac{199}{30}C^2 - \frac{87}{20}\mathcal{E}_4\right)\Sigma_{\mathrm{FV}} + \frac{87}{10}\Sigma_{\mathrm{FV}}\Delta_4\Sigma_{\mathrm{FV}} + \frac{29}{120}R^2\right], \qquad \mathcal{E}_4 = E_4 - \frac{2}{3}\Box R.$$

The independent verifier checks both exact cancellations

$$-\frac{87}{5} + \frac{87}{5} = 0, \qquad \frac{29}{10} - \frac{29}{10} = 0,$$

and the RFT specialization $(199/120, -87/160, 29/120)$.

The decomposition $\Gamma_1[g] = \Gamma_{1,\mathrm{FV}}[g] + W_1[\bar{g}[g]]$ with $R[\bar{g}] = 0$ makes the Ricci-scalar sector dependent. A generic-basis term $RF(\Box)R$ can appear after re-expansion, but its form factor is fixed by the anomaly action and the conformized Weyl sector. It is not a separate coefficient gate in the declared massless conformal FV scope. Derivative-decorated nonlocal cubic-and-higher Weyl form factors, finite normalizations, global inverse data and renormalized products remain open.

6.6 Four-dimensional algebraic cubic Weyl carriers

The next exact receipt isolates the zero-derivative part of the open cubic seam. In four Euclidean dimensions the Weyl tensor is a pair of symmetric tracefree rank-two blocks on self-dual and anti-self-dual two-forms. For three identical blocks, the generator enumerates all fifteen complete pairings of six slots and reduces them under identical-factor permutations:

$$15 \text{ pairings} \longrightarrow 3 \text{ orbits} \longrightarrow 2 \text{ tracefree zeros} + 1 \text{ triangle}.$$

The triangle is $\mathrm{tr}(C_{\pm}^3)$. Orthogonality of the chiral projectors kills cross-block edges, and both mixed allocations then vanish by the trace of the lone block. Thus the parity basis has the exact dimensions

$$\dim V_{C^3}^{\mathrm{even}} = 1, \qquad \dim V_{C^3}^{\mathrm{odd}} = 1,$$

with carriers

$$C_{ab}{}^{cd}C_{cd}{}^{ef}C_{ef}{}^{ab}, \qquad (*C)_{ab}{}^{cd}C_{cd}{}^{ef}C_{ef}{}^{ab}.$$

The verifier independently checks the raw pairing count, every orbit disposition, the exact rank-two sample matrix, the inverse chiral/parity crosswalk, and the even carrier's coordinate in the earlier Schouten-zero Weyl quotient.

This closes only the algebraic carrier subspace. A nonlocal third-curvature functional contains independently labelled $\Box_i$, derivative placements and functions $\Gamma_m(\Box_1, \Box_2, \Box_3)$; unrestricted local integration by parts does not classify that space. The receipt therefore keeps derivative-decorated carriers, form-factor functions and all cubic coefficients at `NOT_COMPUTED`. It also records that the dimension-six C^3 density is not a new dimension-four local one-loop counterterm.

6.7 Parity-even third-curvature carrier quotient

The next receipt imports the complete pure-gravity parity-even conformal carrier list in terms of

$$K_{\mu\nu} = \frac{2}{\Box}\nabla^{\beta}\nabla^{\alpha}C_{\alpha\mu\beta\nu}.$$

The symbol K here denotes this derived nonlocal symmetric tensor, not the four-index Weyl tensor or a kinetic operator. The five rows $I_{10}, I_{24}, I_{25}, I_{28}, I_{29}$ have source-generic stabilizers

$S_3, S_2(23), S_2(23), S_2(12), C_3$, respectively. On the scalar-flat transverse K carrier, I_{29} factorizes in momentum space as

$$-(k_2K_1k_2)(k_3K_2k_3)(k_1K_3k_1).$$

Momentum conservation and $k_iK_i = 0$ equate its two orientations, enhancing its effective stabilizer to S_3. Independent coset enumeration therefore gives effective scalar-flat label dimensions

$$1 + 3 + 3 + 3 + 1 = 11$$

and character $(11, 5, 2)$, hence multiplicities $5\mathbf{1} \oplus 3\mathbf{2}_{\text{std}}$.

The exact four-dimensional identity from CPT-IV (A.35), restricted to the pure-K scalar-flat sector, has one arbitrary completely symmetric form factor. Its coefficient rows contain $I_{10}, I_{24}, I_{25}, I_{28}$ and omit I_{29}. Deleting the trivial S_3 component of I_{28} is one section of the quotient. The resulting character is $(10, 4, 1)$, with multiplicities

$$4\mathbf{1} \oplus 3\mathbf{2}_{\text{std}}.$$

The verifier separately checks every subgroup index, both character decompositions, the symmetry of the coefficient polynomials, and the absence of I_{29} from (A.35). It also evaluates all carriers on an exact 125×11 rational TT fixture, proves the I_{29} reversal identity, finds rank ten and nullity one, and identifies the sole null vector with (A.35). Dependency hashes and fail-closed coefficient flags are checked independently.

This result does not mean that ten analytic form factors have been computed. It means that five carrier-labelled functions, with separately recorded source and effective stabilizers, are subject to one arbitrary symmetric functional relation. The repository functions and coefficients and the parity-odd derivative sector remain at `NOT_COMPUTED`.

6.8 Exact universal CPT source kernels

The next receipt imports the five corresponding third-curvature kernels from the primary CPT source archive and its ancillary symbolic file. It records both source SHA-256 hashes, the source effective-action normalization, the exact rational alpha-parameter numerator, explicit tree and logarithmic terms, the stabilizer symmetrization and the resulting formula-row digest. The common convention is

$$\Gamma_i = \left\langle \frac{df\,f_i}{-\Omega} \right\rangle_3 + T_i + L_i, \qquad \Omega = \alpha_2\alpha_3\Box_1 + \alpha_1\alpha_3\Box_2 + \alpha_1\alpha_2\Box_3.$$

Exact rescaling verifies homogeneities $(-1, -2, -2, -3, -4)$ for $(I_{10}, I_{24}, I_{25}, I_{28}, I_{29})$, and independent parsing checks selected source terms and the full formula digest. The source-generic C_3 provenance of I_{29} is retained, while its raw alpha kernel is independently verified to be fully S_3 symmetric, in agreement with the effective scalar-flat carrier identity above.

The coefficient-bearing source fixture is deliberately narrow: $F = -\Box$ on one real scalar, with $P = 0$ and zero bundle curvature. It is not the conformal-graviton determinant. On tensor and ghost bundles, P and $\mathcal{R}_{\mu\nu}$ are linear in the background curvature and their other CPT rows mix into the same pure-gravity carrier functions. Thus neither constrained Einstein-background ranks nor finitely many local b_4 coordinates identify five functions of three box variables. The receipt fails closed until the repository supplies the same-gauge generic-background full-BV Hessian and complete trace-substitution map through third curvature order, or an equivalent nonminimal fourth-order CPT kernel, with the matching measure.

The production and independent replay commands are

```
PYTHONPATH=quantum-weyl python3 -m \
  transfer.cpt_universal_third_curvature_kernels --check
PYTHONPATH=quantum-weyl python3 -m \
  transfer.verify_cpt_universal_third_curvature_kernels
PYTHONPATH=quantum-weyl python3 -m unittest \
  transfer.tests.test_cpt_universal_third_curvature_kernels
```

6.9 Generic Diff–Weyl ghost matching obstruction

The generic-background gauge rows provide an exact test of whether the minimal scalar source kernels can receive the repository ghost block by a direct trace substitution. Eliminating the algebraic Weyl-ghost row gives

$$(M_{\text{eff}}\xi)_\mu = \Box\xi_\mu + R_{\mu\nu}\xi^\nu + \frac{1}{2}\nabla_\mu\nabla\cdot\xi,$$

independently of the gauge parameter β. The verifier reconstructs the Schur complement at $\beta = 0, 1/4, 1/2$, computes the unit-covector principal spectrum $(3/2, 1, 1, 1)$, and proves a fixed two-sided scalarization no-go from the exact spectrum $(2/3, 3/2, 1, 1)$ of the two-covector comparison matrix. It also applies the curvature term to an explicit tracefree-Ricci jet and finds the transverse mixing vector $2e_1$ from longitudinal input e_0.

The Einstein specialization reproduces $M_{\text{eff}}(\nabla c) = -(3/2)\nabla(\Delta_0 - R/3)c$, so this receipt explains both why the accepted special-background scalar factor is correct and why it does not define the generic determinant. The current minimal-CPT matching architecture is therefore marked `OBSTRUCTED`; the anomaly and QME theorems are unchanged. The production and independent replay commands are

```
PYTHONPATH=quantum-weyl python3 -m \
  spectral.euclidean.generic_background_ghost_cpt_obstruction --check
PYTHONPATH=quantum-weyl python3 -m \
  spectral.euclidean.verify_generic_background_ghost_cpt_obstruction
PYTHONPATH=quantum-weyl python3 -m unittest \
  spectral.euclidean.tests.test_generic_background_ghost_cpt_obstruction
```

6.10 Exact Endo–Duhamel ghost reduction

The architecture obstruction admits a constructive reduction. With $H = -M_{\text{eff}}$, the verifier reconstructs the coefficient triples in the basis $(-\Box\mathbf{1}, \mathrm{Ric}, \nabla\nabla\cdot)$:

$$H = (1, -1, -\tfrac{1}{2}), \quad F = (1, 1, 0), \quad H_0 = (1, 1, -\tfrac{1}{2}), \quad W = (0, -2, 0),$$

and checks $H = H_0 + W$. The version-pinned primary Endo formula specializes at $\alpha = -1/2$ to

$$K_{H_0}(t) = K_F(t) - \nabla\nabla' \int_t^{3t/2} ds\, K_{\Delta_0}(s).$$

The independent replay verifies the flat projector exponents, the longitudinal inverse coefficient $2/3$, the finite proper-time interval, and the nonzero-mode determinant identity. It then constructs the complete cubic work table

$$(n, \text{maximum Endo-kernel order}) = (0, 3), (1, 2), (2, 1), (3, 0).$$

The remaining ghost coefficient calculation is therefore finite, but the listed insertion traces are not silently marked evaluated.

```
PYTHONPATH=quantum-weyl python3 -m \
  spectral.euclidean.generic_background_ghost_endo_duhamel_reduction --check
PYTHONPATH=quantum-weyl python3 -m \
  spectral.euclidean.verify_generic_background_ghost_endo_duhamel_reduction
PYTHONPATH=quantum-weyl python3 -m unittest \
  spectral.euclidean.tests.test_generic_background_ghost_endo_duhamel_reduction
```

6.11 Exact Hodge–resolvent reduction of the one- and two-insertion rows

Integrating the Endo identity on primed nonzero modes and exchanging the proper-time integrals gives the fixed-s interval $2s/3 \le t \le s$. Its width $s/3$, together with $\int_0^\infty sK_{\Delta_0}(s)\,ds = \Delta_0^{-2}$, yields

$$G_{H_0} = G_F - \frac{1}{3} d\Delta_0^{-2}\delta.$$

Substitution in the trace logarithm and exact cyclic reduction leave five minimal carriers:

$$\begin{aligned}
\Gamma^{(1)}_{\mathrm{gh}} &= \mathrm{Tr}_1(G_F W) - \frac{1}{3}\,\mathrm{Tr}_0(\Delta_0^{-2}\delta W d),\\
\Gamma^{(2)}_{\mathrm{gh}} &= -\frac{1}{2}\,\mathrm{Tr}_1(G_F W G_F W) + \frac{1}{3}\,\mathrm{Tr}_0(\Delta_0^{-2}\delta W G_F W d)\\
&\quad - \frac{1}{18}\,\mathrm{Tr}_0(\Delta_0^{-2}\delta W d\,\Delta_0^{-2}\delta W d).
\end{aligned}$$

The independent fixture uses an exact symmetric noncommuting W which mixes the Hodge sectors. It checks $H_0 G_{H_0} = 1$, equality of the direct and reduced traces for $n = 1, 2, 3$, and the cyclic $n = 3$ longitudinal coefficients $(1/3, -1/3, 1/9, -1/81)$.

The reduction is architectural. The pure-vector physical sum is evaluated next; the three rows containing $\delta W d$ require nonminimal principal-symbol kernels.

```
P=spectral.euclidean
M=generic_background_ghost_n1_n2_hodge_resolvent_reduction
PYTHONPATH=quantum-weyl python3 -m "$P.$M" --check
PYTHONPATH=quantum-weyl python3 -m "$P.verify_$M"
PYTHONPATH=quantum-weyl python3 -m unittest "$P.tests.test_$M"
```

6.12 Exact pure-vector one-/two-insertion CPT projection

In the CPT-IV convention $H = \Box + P - R/6$, the scalar-flat positive vector operators $F = -\Box + \mathrm{Ric}$ and $F + W = -\Box - \mathrm{Ric}$ correspond to $P_F = -\mathrm{Ric}$ and $P_H = +\mathrm{Ric}$. Of the P-dependent source rows

$$1, 3, 4, 5, 6, 13, 14, 15, 16, 17, 26,$$

rows $6, 13, 17$ cancel under $P \mapsto -P$, while $4, 5, 15, 16, 26$ vanish by $\mathrm{tr}\, P = R = 0$. Calibration against the independently certified pure-vector $n = 3$ zero-longitudinal sector fixes the positive-determinant/source sign. Thus

$$\Gamma^{(1)}_{\mathrm{vector}} + \Gamma^{(2)}_{\mathrm{vector,vector}} = 6\Gamma_1\mathcal{S}_1 - 2\Gamma_3\mathcal{S}_3 - 2\Gamma_{14}\mathcal{S}_{14}.$$

The source parameter rows are

$$\Gamma_1 = \left\langle \frac{1/3}{\Delta} \right\rangle_3, \quad \Gamma_3 = \left\langle \frac{2\alpha_1\alpha_2}{\Delta} \right\rangle_3, \quad \Gamma_{14} = -\left\langle \frac{2\alpha_3 - 4\alpha_3^2}{x_3\Delta} \right\rangle_3,$$

with

$$\Delta = \alpha_2\alpha_3 x_1 + \alpha_1\alpha_3 x_2 + \alpha_1\alpha_2 x_3.$$

The machine-readable result stores all eleven ordered $I_{10}/I_{24}/I_{25}/I_{28}/I_{29}$ channel integrands. Ten momentum fixtures times 125 TT tensor products and two source structures give 2,500 exact projection identities. Another 750 identities verify the linearized Riemann/Ricci contraction and contracted Bianchi identity. Two unseen fixtures independently replay the coordinates and detect a deliberate coordinate mutation.

The exact missing-carrier theorem is equally important. The other three Hodge rows contain

$$D_W = \delta W d, \qquad \sigma_2(D_W)(p) = W^{\mu\nu} p_\mu p_\nu.$$

This anisotropic curvature-dependent principal symbol is not a minimal bundle endomorphism P, and the mixed row couples scalar and vector resolvents. Minimal CPT P rows cannot evaluate them. Dedicated kernels with one and two D_W insertions, or an equivalent direct nonminimal Endo form-factor calculation, were therefore the smallest missing input before the Schur resummation below.

```
P=spectral.euclidean
M=generic_background_ghost_n1_n2_vector_cpt_projection
PYTHONPATH=quantum-weyl python3 -m "$P.$M" --check
PYTHONPATH=quantum-weyl python3 -m "$P.verify_$M"
PYTHONPATH=quantum-weyl python3 -m unittest "$P.tests.test_$M"
```

6.13 Exact longitudinal Schur resummation

Let $A = F + W$, $H = A + \frac{1}{2} d\delta$, and $H_0 = F + \frac{1}{2} d\delta$ in the differential-form sign convention. The matrix determinant lemma and the Hodge Ward identities give

$$S_L(W) = \frac{2}{3}\mathbf{1} + \frac{1}{3}\delta(F+W)^{-1} d,$$

and, for one compatible determinant-class relative prescription,

$$\mathrm{Det}_{\mathrm{rel}}(H, H_0) = \mathrm{Det}_{\mathrm{rel}}(F+W, F)\,\mathrm{Det}_F(S_L(W), \mathbf{1}).$$

The factor 2/3 removes the base scalar determinant $\mathbf{1} + \frac{1}{2}\delta F^{-1} d = \frac{3}{2}\mathbf{1}$.

With

$$B_n = \Delta_0^{-1} \delta W (G_F W)^{n-1} d \Delta_0^{-1},$$

the exact resolvent series is $S_L = \mathbf{1} + \sum_{n\geq 1} (-1)^n B_n/3$. Expansion of $\mathrm{Tr} \log S_L$ gives

$$-\frac{1}{3}\,\mathrm{Tr}\, B_1,$$

$$\frac{1}{3}\,\mathrm{Tr}\, B_2 - \frac{1}{18}\,\mathrm{Tr}\, B_1^2,$$

and

$$-\frac{1}{3}\,\mathrm{Tr}\, B_3 + \frac{1}{9}\,\mathrm{Tr}(B_1 B_2) - \frac{1}{81}\,\mathrm{Tr}\, B_1^3.$$

The first two rows coincide coefficientwise with the three previously open one-/two-insertion D_W carriers. Thus they are not three independent analytic kernels; all longitudinal insertion orders are one scalar Schur trace-log series. The cubic longitudinal weights are $(-1/3, 1/9, -1/81)$.

The producer checks a noncommuting 4×4 vector/2×2 scalar fixture. The verifier uses an unrelated 5×5 vector/3×3 scalar fixture, replays the determinant and all three trace-log orders, and rejects the mutation $1/9 \mapsto 1/8$. For $\mathrm{Ric} = (R/4)g$, the minimal-vector longitudinal ratio $(\Delta_0 - R/2)/\Delta_0$ times the Schur ratio $(\Delta_0 - R/3)/(\Delta_0 - R/2)$ reproduces the accepted scalar ghost factor.

The regularization boundary is explicit. The finite-dimensional identity and formal trace-log series are exact. Since $S_L - \mathbf{1}$ begins at pseudodifferential order -2 in dimension four, ordinary trace-class Fredholm determinacy does not follow from order counting. The Fredholm identity applies if determinant-class hypotheses hold on a restricted domain; generically a regularized relative determinant or equivalent common trace regulator is required. Separate zeta determinants may also add a local multiplicative anomaly. The selected generic weight-raised local term is evaluated below; the generic Schur relative finite kernel is not, so no repository coefficient or complete ghost form factor is promoted.

```
P=spectral.euclidean
M=generic_background_ghost_longitudinal_schur_resummation
PYTHONPATH=quantum-weyl python3 -m "$P.$M" --check
PYTHONPATH=quantum-weyl python3 -m "$P.verify_$M"
PYTHONPATH=quantum-weyl pytest -q \
 quantum-weyl/spectral/euclidean/tests/test_$M.py
```

6.14 Sharp Schatten split and critical Schur residue

Write $S_L = \mathbf{1} + K$. The exact resolvent identity gives

$$K = -\frac{1}{3}\delta(F+W)^{-1}W d\Delta_0^{-1}, \qquad \sigma_{-2}(K) = -\frac{1}{3}\frac{\langle \xi, W\xi\rangle}{|\xi|^4}.$$

On a closed compact four-manifold, $K \in \mathcal{S}_p$ for every $p > 2$. Thus $K \in \mathcal{S}_3$, K^3 is trace class, and

$$\det{}_3(\mathbf{1}+K) = \det\Big[(\mathbf{1}+K)e^{-K+K^2/2}\Big]$$

is a genuine modified Fredholm determinant. Its small-norm series begins at cubic order:

$$\log\det{}_3(\mathbf{1}+K) = \sum_{m\geq 3}\frac{(-1)^{m+1}}{m}\operatorname{Tr}K^m.$$

For one declared trace extension $\mathcal{R}$, the complete bookkeeping split is therefore

$$\log\mathrm{Det}_{3,\mathcal{R}}(\mathbf{1}+K) = \mathcal{R}(K) - \frac{1}{2}\mathcal{R}(K^2) + \log\det{}_3(\mathbf{1}+K).$$

With $K_1 = -B_1/3$, $K_2 = B_2/3$, and $K_3 = -B_3/3$, this reproduces

$$\begin{aligned}
O(W) &: -\tfrac{1}{3}\mathcal{R}(B_1),\\
O(W^2) &: \tfrac{1}{3}\mathcal{R}(B_2) - \tfrac{1}{18}\mathcal{R}(B_1^2),\\
O(W^3) &: -\tfrac{1}{3}\mathcal{R}(B_3) + \tfrac{1}{9}\mathcal{R}(B_1B_2) - \tfrac{1}{81}\operatorname{Tr}B_1^3.
\end{aligned}$$

Only the last cubic term is already in the canonical $\det_3$ tail.

The order--4 symbol of K^2 is simply

$$\sigma_{-4}(K^2) = \frac{1}{9}\frac{\langle \xi, W\xi\rangle^2}{|\xi|^8}.$$

Using $\operatorname{vol} S^3 = 2\pi^2$ and

$$\langle n_i n_j n_k n_l \rangle_{S^3} = \frac{\delta_{ij}\delta_{kl} + \delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}}{24}$$

gives

$$\operatorname{Wres}(K^2) = \frac{1}{(4\pi)^2}\int \frac{(\operatorname{tr} W)^2 + 2\operatorname{tr}(W^2)}{108} = \frac{1}{(4\pi)^2}\int \frac{R^2 + 2R_{\mu\nu}R^{\mu\nu}}{27}.$$

The producer performs this contraction symbolically for a generic symmetric endomorphism. The verifier uses a different fixed endomorphism, integrates its degree-four sphere polynomial monomial by monomial, and rejects $108 \mapsto 109$. A separate noncommuting 4×4 fixture verifies the $\det_3$ series through sixth order.

The remaining local residue is obtained without expanding the full symbol recursively. Through order -4,

$$K = -\frac{1}{3}B_1 + \frac{1}{3}B_2 + O(\Psi^{-6}), \quad B_1 = \Delta_0^{-1}\delta W d\Delta_0^{-1}, \quad B_2 = \Delta_0^{-1}\delta W G_F W d\Delta_0^{-1}.$$

Cyclicity reduces B_1 to the heat insertion $(\delta W d)\Delta_0^{-2}$. The mixed scalar kernel and the principal symbol of B_2 give

$$\begin{aligned}
\operatorname{Wres}(B_1) &= \frac{1}{(4\pi)^2}\int \left[-\frac{1}{3}W^{\mu\nu}R_{\mu\nu} + \frac{1}{6}(\operatorname{tr} W)R\right],\\
\operatorname{Wres}(B_2) &= \frac{1}{(4\pi)^2}\int \frac{1}{2}\operatorname{tr}(W^2).
\end{aligned}$$

Consequently, for $W = -2\operatorname{Ric}$,

$$\operatorname{Wres}(K) = \frac{1}{(4\pi)^2}\int \frac{R^2 + 4R_{\mu\nu}R^{\mu\nu}}{9}, \qquad \operatorname{Wres}\log S_L = \frac{1}{(4\pi)^2}\int \frac{5R^2 + 22R_{\mu\nu}R^{\mu\nu}}{54}.$$

The latter uses $\operatorname{Wres}\log(1+K) = \operatorname{Wres}(K) - \frac{1}{2}\operatorname{Wres}(K^2)$ because every higher power has order below -4. The independent verifier reconstructs both coefficients from the mixed heat row and the sphere second moment. It also replays the exact Einstein ratio and the isotropic test $W = w\mathbf{1}$, and rejects a mutated mixed-kernel Ricci coefficient.

Choose the positive order-two scalar weight

$$Q = \Delta_0 + \Pi_0, \qquad Q_\mu = Q/\mu^2, qquad \mathcal{R}_\mu(A) = \operatorname*{FP}_{z=0}\operatorname{TR}'(AQ_\mu^{-z}).$$

The smoothing projector changes no residue. The weighted-trace Laurent row and the square on the dimensionful scale give

$$\operatorname{TR}'(AQ^{-z}) = \frac{\operatorname{Wres}(A)}{2z} + \mathcal{R}_Q(A) + O(z), qquad \mathcal{R}_{e^t\mu}(A) - \mathcal{R}_\mu(A) = t\operatorname{Wres}(A).$$

Thus the exact Schur pole and scale rows are

$$\operatorname*{Res}_{z=0}\operatorname{TR}'((\log S_L)Q^{-z}) = \frac{1}{(4\pi)^2}\int \frac{5R^2 + 22\operatorname{Ric}^2}{108},$$

$$\frac{\mathrm{d}}{\mathrm{d}\log\mu}\log\operatorname{Det}_{3,\mathcal{R}_\mu}(S_L)=\frac{1}{(4\pi)^2}\int\frac{5R^2+22\,\mathrm{Ric}^2}{54}.$$

The independent verifier reconstructs the factor 'scale power / weight order = 2/2' and rejects the mutated choice Q/μ. The generic-background reference-scale finite $\mathcal{R}_{\mu_0}(K)$ row and the finite part of $\mathcal{R}_{\mu_0}(K^2)$ remain open. They are not determined by these local residue data. The selected generic weight-raised noncommuting multiplicative term is fixed below because it is local.

On the round unit S^4, however, the complete scalar spectrum supplies an exact benchmark. After deleting the absent $\ell=0$ gradient and the five $\ell=1$ proper-conformal-Killing ghost zero modes,

$$\lambda_\ell=\ell(\ell+3),\qquad d_\ell=\frac{(2\ell+3)(\ell+2)(\ell+1)}{6},\qquad K_\ell=\frac{2}{\lambda_\ell-6},\qquad \ell\geq 2.$$

Write $u_\pm=(7\pm\sqrt{33})/2$, $\Psi=\psi(u_-)+\psi(u_+)$ and $\Psi_1=\psi^{(1)}(u_-)-\psi^{(1)}(u_+)$. At $\mu_0=1$ in inverse-radius units, exact Hurwitz-zeta continuation and the $B=\Delta_0-6$ to $Q=\Delta_0$ weight change give

$$\mathcal{R}_{\Delta_0}(K)=-\frac{20}{9}-\frac{8}{3}\Psi,\qquad \operatorname{FP}\mathcal{R}_{\Delta_0}(K^2)=-\frac{2}{3}\Psi+\frac{16}{3\sqrt{33}}\Psi_1.$$

The independent verifier reconstructs the finite parts from a convergent Hurwitz-zeta binomial series and obtains

$$-3.0967576144286354\ldots,qquad2.7591028732128106\ldots.$$

It also checks the exact weight shift $(-2,0)$ and the five-dimensional $\ell=1$ kernel. The remaining trace-class series is bounded without a floating-point tail assumption. Even alternating-log bounds cover $2\leq\ell\leq 20$; the rest is reduced to positive Hurwitz-zeta tails whose Euler–Maclaurin remainders are bounded in exact rational arithmetic. This gives

$$\log\det{}_3(I+K)=0.49816356541962909843125329994148187238611929 34\ldots$$

with interval width below 5.8×10^{-48}, and therefore

$$\log\operatorname{Det}_{3,\mathcal{R}_{\Delta_0}}(S_L)=-3.9781454856154116274753955548059869205821661933\ldots.$$

An independent direct infinite-mode summation lies strictly inside the rational enclosure. A rank-one smoothing mutation changes the two finite rows by 7/11 and 126/121 without changing any homogeneous symbol or residue. This proves that the generic gate needs a full primed Green kernel or spectral measure rather than another local heat coefficient.

The commuting round-sphere factorization has one further exact local row. On the primed $\ell\geq2$ carrier let $Q=\Delta_0$, $A=\Delta_0-4$ and $B=\Delta_0-6$, so $S_L=AB^{-1}$. The zeta-to-weighted determinant comparison gives

$$m_Q(A,B)=-\frac{1}{4}(4^2-6^2)\,\mathrm{Wres}(Q^{-2}),\qquad \mathrm{Wres}(Q^{-2})=\frac{1}{3},$$

and hence

$$m_Q(A,B)=\frac{5}{3}.$$

Combining this with the weighted modified determinant yields

$$\log\det_\zeta A-\log\det_\zeta B=-2.3114788189487449608087288881393202539\ldots.$$

The independent verifier reconstructs the same value from 320 exact Hurwitz-zeta continuation terms and rejects mutations of the residue, shift pair, zero-mode policy, or generic-background claim flag. This special commuting result uses an Einstein numerator/denominator factorization. It is not the generic weight-raised prescription computed next.

For the generic carrier, no canonical numerator/denominator pair exists. The already frozen weight instead selects

$$A = S_L Q, \qquad B = Q, \qquad Q = \Delta_0 + \Pi_0,$$

with the left-Schur ordering and a common Agmon sector on one connected admissible stratum. If $X = \log S_L \in \Psi^{-2}$ and $Y = \log Q$, then

$$\log(S_L Q) - X - Y = \frac{1}{2}[X, Y] + \frac{1}{12}[Y, [Y, X]] \pmod{\Psi^{-5}}.$$

The two displayed rows have orders -3 and -4; the next independent row has order -6. Applying the Q-weighted commutator defect with $Y = \log Q$ makes both surviving commutator traces vanish. Cross terms with X lie below residue order and $(\log(1 + K))^2 = K^2 \pmod{\Psi^{-6}}$. Consequently

$$\boxed{m_Q^{\mathrm{wr}}(S_L) = \log \det_\zeta(S_L Q) - \log \det_\zeta Q - \operatorname{tr}^Q \log S_L = -\frac{1}{4} \operatorname{Wres}(K^2)}$$

and hence

$$m_Q^{\mathrm{wr}}(S_L) = -\frac{1}{(4\pi)^2} \int \frac{R^2 + 2\operatorname{Ric}^2}{108}.$$

On round unit S^4 the residue is $4/3$, the defect is $-1/3$, and the zeta ratio is

$$-4.31147881894874496080872888813932025 39\ldots.$$

An independent Hurwitz-zeta continuation replays the last number. Its difference from the preceding $5/3$ defect is exactly 2 because the factorization conventions differ. This closes the selected generic local BCH row while leaving the global generic finite rows open.

```
P=spectral.euclidean
M=generic_background_ghost_schur_schatten_split
PYTHONPATH=quantum-weyl python3 -m "$P.$M" --check
PYTHONPATH=quantum-weyl python3 -m "$P.verify_$M"
PYTHONPATH=quantum-weyl pytest -q \
 quantum-weyl/spectral/euclidean/tests/test_$M.py

M=generic_background_ghost_schur_wodzicki_residue
PYTHONPATH=quantum-weyl python3 -m "$P.$M" --check
PYTHONPATH=quantum-weyl python3 -m "$P.verify_$M"
PYTHONPATH=quantum-weyl pytest -q \
 quantum-weyl/spectral/euclidean/tests/test_$M.py

M=generic_background_ghost_schur_weighted_trace_scale
PYTHONPATH=quantum-weyl python3 -m "$P.$M" --check
PYTHONPATH=quantum-weyl python3 -m "$P.verify_$M"
PYTHONPATH=quantum-weyl pytest -q \
 quantum-weyl/spectral/euclidean/tests/test_$M.py

M=round_s4_ghost_schur_finite_weighted_traces
```

```
PYTHONPATH=quantum-weyl python3 -m "$P.$M" --check
PYTHONPATH=quantum-weyl python3 -m "$P.verify_$M"
PYTHONPATH=quantum-weyl pytest -q \
 quantum-weyl/spectral/euclidean/tests/test_$M.py

M=round_s4_ghost_schur_zeta_factorization
PYTHONPATH=quantum-weyl python3 -m "$P.$M" --check
PYTHONPATH=quantum-weyl python3 -m "$P.verify_$M"
PYTHONPATH=quantum-weyl pytest -q \
 quantum-weyl/spectral/euclidean/tests/test_$M.py

M=generic_background_ghost_schur_weight_raised_zeta_factorization
PYTHONPATH=quantum-weyl python3 -m "$P.$M" --check
PYTHONPATH=quantum-weyl python3 -m "$P.verify_$M"
PYTHONPATH=quantum-weyl pytest -q \
 quantum-weyl/spectral/euclidean/tests/test_$M.py
```

6.15 Exact three-insertion adiabatic angular carrier

At cubic curvature order the three-insertion row uses only the flat Endo inverse

$$G_0(p) = \frac{1}{p^2} P(n), \qquad P(n) = I - \frac{1}{3} n n^T, \qquad n = \frac{p}{|p|}.$$

For a symmetric Ricci endomorphism R, the independent verifier enumerates all pairings in the second, fourth, and sixth isotropic sphere moments and obtains

$$\langle \mathrm{tr}(PRPRPR) \rangle_{S^3} = \frac{503}{648} \,\mathrm{tr}\, R^3 + \frac{11}{864} (\mathrm{tr}\, R)(\mathrm{tr}\, R^2) - \frac{1}{5184} (\mathrm{tr}\, R)^3.$$

Multiplication by $W^3 = (-2R)^3$ and by the $1/3$ coefficient in $\mathrm{Tr} \log(1 + G_0 W)$ gives

$$J_3 \left[-\frac{503}{243} \,\mathrm{tr}\, R^3 - \frac{11}{324} (\mathrm{tr}\, R)(\mathrm{tr}\, R^2) + \frac{1}{1944} (\mathrm{tr}\, R)^3 \right], \qquad J_3 = \int \frac{d^4 p}{(2\pi)^4} \frac{1}{(p^2)^3}.$$

On the scalar-flat carrier only the first term remains. This is an exact tensor numerator, not the full triangle form factor: at zero external momentum J_3 is scaleless and infrared singular. The nonzero-momentum triangle and its repository projection are certified separately; the three longitudinal D_W one-/two-insertion towers are resummed by the preceding Schur factor. Its K, K^2, and logarithmic residues and the declared order-two weighted-trace pole/scale row are exact; the round-S^4 finite rows, canonical $\det_3$ tail, weighted modified determinant, and zeta-to-weighted defect $5/3$ are complete. Their generic finite versions require a full primed Green/spectral carrier. For the selected generic weight-raised convention the noncommuting BCH trace and local $-\frac{1}{4}\,\mathrm{Wres}(K^2)$ term are exact. Polarizing the cubic polynomial gives the complete zero-momentum S_3 carrier on (R_1, R_2, R_3): its three basis elements are the symmetrized cyclic trace, the cyclic sum of one trace times one pair trace, and the product of three traces, with the same three coefficients shown above. The verifier checks all six permutations on noncommuting symmetric matrices and recovers the diagonal polynomial.

```
PYTHONPATH=quantum-weyl python3 -m \
  spectral.euclidean.generic_background_ghost_n3_adiabatic_carrier --check
```

```
PYTHONPATH=quantum-weyl python3 -m \
  spectral.euclidean.verify_generic_background_ghost_n3_adiabatic_carrier
PYTHONPATH=quantum-weyl python3 -m unittest \
  spectral.euclidean.tests.test_generic_background_ghost_n3_adiabatic_carrier
```

6.16 Exact nonzero-momentum three-insertion parametric kernel

For generic nonexceptional Euclidean momenta $k_1 + k_2 + k_3 = 0$, route $q_0 = p$, $q_1 = p + k_1$, and $q_2 = p - k_3$. The complete labelled-Ricci integrand is

$$\frac{\mathrm{tr}[\Pi(q_0)R_1\Pi(q_1)R_2\Pi(q_2)R_3]}{q_0^2 q_1^2 q_2^2}, \qquad \Pi(q) = I - \frac{1}{3}\frac{qq^T}{q^2}.$$

Expansion of the three longitudinal projectors gives exactly eight sectors, with multiplicities $(1, 3, 3, 1)$. If a sector contains s longitudinal projectors, Feynman parametrization and the shifted four-dimensional loop integral give rows $m = 0, \ldots, s$ with coefficient and denominator

$$c_{s,m} = \frac{(s-m)!}{2^m}, \qquad \Delta^{m-s-1},$$

where

$$\Delta = \alpha_0\alpha_1 k_1^2 + \alpha_1\alpha_2 k_2^2 + \alpha_2\alpha_0 k_3^2, \qquad \alpha_0 + \alpha_1 + \alpha_2 = 1.$$

There are twenty rational Wick rows in total. The combined W^3 and $\mathrm{Tr}\log$ multiplier is

$$-\frac{8}{3}(4\pi)^{-2}.$$

An independent exact replay checks the direct eight-sector integrand and cyclic covariance on noncommuting symmetric endomorphisms. This closes the generic labelled- Ricci parametric triangle before its separately certified repository projection. The zero-derivative sector can feed I_{10}, while the longitudinal sectors carry two, four, and six external derivatives and can feed $I_{24}, I_{25}, I_{28}, I_{29}$. The three longitudinal D_W one-/two-insertion towers are resummed by the preceding Schur factor; its K, K^2, and logarithmic residues and the declared order-two weighted-trace pole/scale row are fixed. On round S^4 the two finite trace rows, canonical $\det_3$ tail, weighted modified determinant, and zeta-to-weighted defect $5/3$ are complete. Their generic finite versions and the physical fourth-order carrier remain unevaluated; the selected generic weight-raised BCH residue is exact.

```
PYTHONPATH=quantum-weyl python3 -m \
  spectral.euclidean.generic_background_ghost_n3_triangle_kernel --check
PYTHONPATH=quantum-weyl python3 -m \
  spectral.euclidean.verify_generic_background_ghost_n3_triangle_kernel
PYTHONPATH=quantum-weyl python3 -m unittest \
  spectral.euclidean.tests.test_generic_background_ghost_n3_triangle_kernel
```

6.17 Scalar-flat K–Ricci normalization crosswalk

The carrier convention is

$$K_{\mu\nu} = \frac{2}{\Box}\nabla^\beta\nabla^\alpha C_{\alpha\mu\beta\nu}.$$

In four dimensions, the contracted Weyl identity contributes the coefficient $(d-3)/(d-2) = 1/2$. On the declared scalar-flat source complement, the contracted Bianchi identity and $[\nabla, \nabla]\,\mathrm{Ric} = O(\mathcal{R}^2)$ therefore give

$$\nabla^\beta \nabla^\alpha C_{\alpha\mu\beta\nu} = \frac{1}{2}\Box R_{\mu\nu} + O(\mathcal{R}^2), \qquad K_{\mu\nu} = R_{\mu\nu} + O(\mathcal{R}^2).$$

Replacing one of three linear K factors by the remainder first contributes at order $2 + 1 + 1 = 4$. Thus the labelled cubic K product and labelled cubic Ricci product agree through $O(\mathcal{R}^3)$. An independent exact fixture with $k = (1, 0, 0, 0)$ and a nontrivial TT polarization verifies componentwise $k^\alpha k^\beta C_{\alpha\mu\beta\nu} = (1/2)k^2 R_{\mu\nu}$, guarding the sign and index order.

This receipt closes the normalization seam. Derivative counting maps the zero-, one-, two- and three-longitudinal-projector sectors respectively to possible derivative orders (0), $(0, 2)$, $(0, 2, 4)$ and $(0, 2, 4, 6)$. Consequently the complete target is $(I_{10}, I_{24}, I_{25}, I_{28}, I_{29})$. The next subsection performs the parametric tensor/form-factor projection.

```
PYTHONPATH=quantum-weyl python3 -m \
  transfer.scalar_flat_k_ricci_crosswalk --check
PYTHONPATH=quantum-weyl python3 -m \
  transfer.verify_scalar_flat_k_ricci_crosswalk
PYTHONPATH=quantum-weyl python3 -m unittest \
  transfer.tests.test_scalar_flat_k_ricci_crosswalk
```

6.18 Exact ghost $n = 3$ five-carrier projection

The five carrier labels have eleven effective labelled orientations with multiplicities $(1, 3, 3, 3, 1)$. On each of ten generic momentum fixtures the producer constructs five exact TT polarizations on every leg and evaluates all $5^3 = 125$ tensor products. The resulting carrier matrix has shape 125×11 and exact rank ten. The appended section row

$$(0, \ldots, 0, 1, 1, 1, 0)$$

imposes $\Gamma^{123}_{28} + \Gamma^{132}_{28} + \Gamma^{231}_{28} = 0$ and completes the rank to eleven. This fixes a representative modulo the unique arbitrary symmetric CPT-IV relation; it does not add an independent identity.

The exact triangle solve retains (α_1, α_2) symbolically. After multiplication by the common Δ^4, each channel numerator is a rational polynomial. Its box degree is $3 - d/2$ for explicit derivative order $d = 0, 2, 4, 6$, and its alpha degree is at most nine. Homogeneous box interpolation uses ten unisolvent fixtures, enough for the maximum degree- three sector. The stored row counts are

$$(195;\ 65, 103, 65;\ 97, 91, 97;\ 37, 40, 37;\ 10),$$

for a total of 837 nonzero rational terms.

The independent fast verifier evaluates the stored formula on two momentum fixtures absent from interpolation and two rational simplex points absent from generation. At each fixture it compares the reconstructed carrier combination against all 125 direct Endo-triangle tensor amplitudes. Every exact residual and the symmetric I_{28} coordinate vanish. It also rejects a changed coefficient after recomputing the formula digest. Full symbolic regeneration takes about 163 seconds and is assigned to the scientific/ release tier; the unseen exact replay takes about four seconds.

```
PYTHONPATH=quantum-weyl python3 -m \
  spectral.euclidean.verify_generic_background_ghost_n3_five_carrier_projection
```

```
TEST="spectral.euclidean.tests."
TEST="${TEST}test_generic_background_ghost_n3_five_carrier_projection"
PYTHONPATH=quantum-weyl python3 -m unittest "$TEST"
```

The result is the parametric $n = 3$ ghost contribution in the declared scalar-flat quotient section. The generic kinematic simplex integrals, normalized longitudinal Schur $\mathcal{R}(K)$ and finite $\mathcal{R}(K^2)$ rows, physical fourth-order Hessian kernel, and their complete sum into five repository functions and coefficients remain open.

6.19 Generic barycentric factorization

The independent generic reduction substitutes $C = 1 - A - B$ only after testing divisibility by

$$\Delta = CAx_1 + ABx_2 + BCx_3.$$

For ten rows the polynomial remainder after division by Δ vanishes exactly. Rehomogenizing their quotients to alpha degree seven gives pole- three integrands. The remaining I_{29} numerator rehomogenizes directly to $-16A^3B^3C^3/27$ at pole four.

The certificate stores every reduced rational coefficient and independently reconstructs all 837 upstream terms. Monomial valuations then give the exact edge and vertex orders printed in the main paper. The current integrands are locally integrable because every $v - p + 2$ is positive, but this does not settle the flux of a future primitive. The three pole-three I_{28} rows reduce to

$$\begin{aligned}
\widetilde{N}_{28}^{123} &= \frac{32}{243}A^2B^2C^2(2A - B - C), \\
\widetilde{N}_{28}^{132} &= -\frac{32}{243}A^2B^2C^2(A + B - 2C), \\
\widetilde{N}_{28}^{231} &= -\frac{32}{243}A^2B^2C^2(A - 2B + C),
\end{aligned}$$

so the quotient relation is a pointwise polynomial identity.

```
MOD="spectral.euclidean.verify_generic_background_ghost_n3_"
MOD="${MOD}barycentric_factorization"
PYTHONPATH=quantum-weyl python3 -m "$MOD"
TEST="spectral.euclidean.tests.test_generic_background_ghost_n3_"
TEST="${TEST}barycentric_factorization"
PYTHONPATH=quantum-weyl python3 -m unittest "$TEST"
```

The next calculation must solve for relative rational IBP primitives. It must record their open-edge and punctured-corner flux separately and reduce the result to the scalar triangle and edge-bubble masters. No generic integrated form factor is claimed by the present factorization.

6.20 Exact symmetric-point simplex integration

At $x_1 = x_2 = x_3 = 1$, permutation averaging preserves every simplex integral and reduces the eleven stored numerators to symmetric polynomials in

$$e_2 = \alpha_0\alpha_1 + \alpha_1\alpha_2 + \alpha_2\alpha_0, \qquad e_3 = \alpha_0\alpha_1\alpha_2.$$

The only transcendental master is

$$J_\triangle = \int_{\Delta_2} \frac{\mathrm{d}^2\alpha}{e_2} = \frac{4}{\sqrt{3}}\,\mathrm{Cl}_2\Big(\frac{\pi}{3}\Big) = 2.3439072386894588906015622888722 77\ldots.$$

Four exact integration-by-parts identities give

$$\int \frac{e_3}{e_2} = \frac{11-4J_\triangle}{54}, \qquad \int \frac{e_3}{e_2^2} = \frac{J_\triangle - 2}{3},$$
$$\int \frac{e_3^2}{e_2^3} = \frac{10J_\triangle - 23}{54}, \qquad \int \frac{e_3^3}{e_2^4} = \frac{62J_\triangle - 145}{486}.$$

For each row the receipt stores the rational polynomial defining the divergence vector field, verifies the divergence identity exactly, and records both open-edge and punctured-corner flux vanishing. Substitution in the symmetrized carrier numerators gives the eleven values printed in the main paper. The independent verifier reconstructs those numerators from the upstream 837-term projection, checks every rational identity, and separately quadratures the four smooth sector integrals and the one-dimensional Clausen master. This second path is important: direct iterated symbolic integration of the unsymmetrized functions can choose inconsistent logarithm branches at the moving simplex endpoint.

```
MOD="spectral.euclidean.verify_generic_background_ghost_n3_"
MOD="${MOD}symmetric_point_simplex_integration"
PYTHONPATH=quantum-weyl python3 -m "$MOD"
TEST="spectral.euclidean.tests.test_generic_background_ghost_n3_"
TEST="${TEST}symmetric_point_simplex_integration"
PYTHONPATH=quantum-weyl python3 -m unittest "$TEST"
```

This closes only the normalized symmetric-point ghost $n = 3$ fixture. The generic functions, complete ghost determinant, physical fourth-order Hessian, combined repository coefficients and complete Γ_1/Q_1 remain open.

6.21 Exact local BoxR scheme conversion

The zeta/proper-time calculation imports the published local heat-kernel rows for the traceless tensor, gauge-weight and diffeomorphism-ghost operators. An independent exact replay of their $(2, -1, -2)$ determinant combination gives

$$b_\Box^{\mathrm{raw}} = \frac{7}{2}\log\frac{3}{2} - \frac{159}{80} < 0.$$

The repository primitive $\omega\Box R = -(1/12)s(R^2)$ modulo d then fixes

$$z_R^{\mathrm{raw}\to 0} = \frac{7}{24}\log\frac{3}{2} - \frac{53}{320}.$$

The verifier checks that $b_\Box^{\mathrm{raw}} - 12z_R^{\mathrm{raw}\to 0} = 0$ and that adding the same shift to $391/960 - (7/24)\log(3/2)$ gives the stored local coefficient $29/120$. Alternating-series rational bounds certify the strict sign without floating-point arithmetic. The receipt keeps the strict-metric local scheme counterterm separate from the independent dressed $R(\widehat{g})^2$ class in the compensator theory. The subsequent FV certificate, not the local scheme conversion alone, proves that generic-basis nonlocal Ricci terms are structurally determined.

The same certificate binds the contraction non-merge: the Berger ‘tau‘ row is the temporal diffeomorphism ghost in the positive-Berger clock complex, whereas the Wess–Zumino ‘tau‘ row obeys $Q\tau = \mathcal{L}_\xi\tau + \omega$. The identical printed symbol is not a cross-background carrier map.

7 Independent verifiers and mutation tests

The producer and verifier paths are distinct. The verifiers independently:

- validate strict Draft 2020–12 schemas;
- replay exact matrix ranks and contraction identities;
- compare generated certificates with fresh deterministic builds;
- reject deletion of the dressed R^2 class;
- reject authorization of residual transfer;
- reject promotion of the fixed Wess–Zumino contribution to a complete renormalized Q_1;
- replay the diagonal Paneitz/Riegert response solve and reject promotion of a generalized inverse to an unconditional global inverse;
- replay the flat-TT logarithmic coefficient, scale response and momentum difference, and reject promotion of its additive constant or curved completion;
- replay the curvature-order filtration for the covariant logarithm and reject promotion of the cubic remainder, finite constants or Berger/WZ contraction merge;
- replay the Fradkin–Vilkovisky scalar-flat and Weyl-weight identities, and reject identification of its nonlocal representative with the local τ-adic dressed metric or promotion to derivative-decorated nonlocal cubic form factors;
- replay the exact FV anomaly-action response cancellations and RFT specialization, while rejecting a separate nonlocal R^2 datum or a complete Γ_1;
- replay the raw $\Box R$ heat-kernel arithmetic, its exact strict-metric R^2 scheme shift and the 29/120 cross-check, while rejecting promotion to an absolute dressed normalization;
- replay the five third-curvature carrier stabilizers, the raw and four-dimensional quotient characters, and reject promotion of carrier labels to repository form-factor functions or coefficients;
- replay the generic Diff–Weyl ghost Schur complement, nonminimal principal spectrum and tracefree-Ricci Hodge mixing, while rejecting direct minimal-CPT substitution and any promotion to a completed ghost determinant;
- replay the exact Endo split, finite-proper-time heat kernel, nonzero-mode determinant identity and cubic Duhamel insertion bound, while rejecting promotion of the unevaluated insertion traces;
- integrate the Endo identity to the exact Hodge resolvent, reduce the $n = 1, 2$ rows to five minimal vector/scalar carriers, and reject promotion of that architecture to evaluated form factors or coefficients;
- import CPT rows $1, 3, 14$, project the combined pure-vector $n = 1 + n = 2$ slice, replay unseen fixtures, and reject promotion across the three longitudinal D_W carriers;

- replay the longitudinal Schur determinant on a distinct noncommuting fixture through cubic order, detect a mixed-cubic mutation, and reject both evaluation of its generic relative finite kernel and an undeclared change of zeta-factorization convention;
- compute the commuting round-S^4 zeta-to-weighted defect 5/3 from $\mathrm{Wres}(\Delta_0^{-2}) = 1/3$, replay the determinant ratio by independent Hurwitz-zeta continuation, then compute the distinct generic weight-raised local defect $-\frac{1}{4}\,\mathrm{Wres}(K^2)$ and its round-S^4 value $-1/3$ by an independent spectral continuation;
- prove the sharp $\mathcal{S}_3$ trace-ideal disposition, replay the modified Fredholm series through sixth order, compute the critical $\mathrm{Wres}(K^2)$ coefficient by two independent sphere-moment calculations, and reject promotion to the regulated $\mathcal{R}(K)$ or finite $\mathcal{R}(K^2)$ rows;
- enumerate the second, fourth, and sixth isotropic sphere moments for the three-insertion adiabatic angular carrier, while rejecting promotion of its IR-singular radial integral to a nonzero-momentum triangle or I_{10} map;
- replay the eight projector sectors and twenty exact simplex/Wick rows of the generic nonzero-momentum three-insertion kernel, while rejecting promotion of the labelled-Ricci kernel to the repository five-carrier form factors;
- replay the contracted-Weyl coefficient, direct TT sign fixture and fourth-order first replacement error in the scalar-flat K–Ricci crosswalk, while rejecting promotion of normalization alone to a five-carrier projection;
- replay the stored eleven-to-ten five-carrier projection on unseen exact momentum/simplex fixtures across all 125 TT tensor products, and reject a coefficient mutation even after its formula digest is recomputed;
- divide every generic projected numerator by Δ, reconstruct the ten exact cancellations and one pole-four remainder, verify every homogeneous edge/vertex order and the pointwise I_{28} relation, while rejecting any promotion to a completed relative-IBP reduction;
- reconstruct the symmetric-point numerators from the upstream 837 terms, verify the four rational divergence identities and all boundary-flux orders, and compare the exact Clausen reduction with independent quadrature;
- reject changes to the cotangent signs, gradings, or quartet identity;
- validate every content hash used by the paper claim map.

The generated tables add a publication-layer drift guard. Their verifier loads all authoritative table inputs, checks the exact extended boundary matrix and coefficient image, rebuilds the TeX byte-for-byte, and rejects a stale include.

8 Reproduction commands

Run from the repository root:

```
PYTHONPATH=quantum-weyl python3 -m \
  anomalies.wess_zumino_minimal_bv_cotangent_lift --check
PYTHONPATH=quantum-weyl python3 -m \
```

```
  anomalies.verify_wess_zumino_minimal_bv_cotangent_lift
PYTHONPATH=quantum-weyl python3 -m \
  anomalies.wess_zumino_extended_local_bv_cohomology --check
PYTHONPATH=quantum-weyl python3 -m \
  anomalies.verify_wess_zumino_extended_local_bv_cohomology
PYTHONPATH=quantum-weyl python3 -m \
  transfer.one_loop_slavnov_q1_disposition --check
PYTHONPATH=quantum-weyl python3 -m \
  transfer.verify_one_loop_slavnov_q1_disposition
PYTHONPATH=quantum-weyl python3 -m \
  transfer.flat_tt_logarithmic_gamma1 --check
PYTHONPATH=quantum-weyl python3 -m \
  transfer.verify_flat_tt_logarithmic_gamma1
PYTHONPATH=quantum-weyl python3 -m \
  transfer.curvature_squared_covariant_log_gamma1 --check
PYTHONPATH=quantum-weyl python3 -m \
  transfer.verify_curvature_squared_covariant_log_gamma1
PYTHONPATH=quantum-weyl python3 -m \
  transfer.fv_conformized_c2_log_gamma1 --check
PYTHONPATH=quantum-weyl python3 -m \
  transfer.verify_fv_conformized_c2_log_gamma1
PYTHONPATH=quantum-weyl python3 -m \
  transfer.fv_anomaly_action_ricci_sector --check
PYTHONPATH=quantum-weyl python3 -m \
  transfer.verify_fv_anomaly_action_ricci_sector
PYTHONPATH=quantum-weyl python3 -m \
  transfer.verify_third_curvature_weyl_manifest
PYTHONPATH=quantum-weyl python3 -m \
  transfer.verify_cpt_universal_third_curvature_kernels
PYTHONPATH=quantum-weyl python3 -m \
  transfer.verify_cubic_weyl_carrier_basis
PYTHONPATH=quantum-weyl python3 -m \
  spectral.euclidean.verify_generic_background_ghost_n3_five_carrier_projection
PYTHONPATH=quantum-weyl python3 -m \
  spectral.euclidean.verify_generic_background_ghost_longitudinal_schur_resummation
PYTHONPATH=quantum-weyl python3 -m \
  spectral.euclidean.verify_box_r_scheme_conversion
python3 paper/generate_12_quantum_anomaly_tables.py --check
python3 paper/verify_12_quantum_anomaly_tables.py
python3 paper/generate_12_pure_weyl_one_loop_bv_anomaly_claim_map.py --check
python3 paper/verify_12_pure_weyl_one_loop_bv_anomaly_claim_map.py
```

The intentionally slow repository-wide suite is not required for this scoped artifact. The relevant exact unit modules, independent verifiers, atlas validator, and two-pass LaTeX build form the declared publication tier.

9 Remaining gate

The next theorem requires actual new physical input:

1. a same-background compensator-inclusive classical contraction $(\iota_{\mathrm{ext}}, \pi_{\mathrm{ext}}, S_{\mathrm{ext}})$;
2. evaluation of the generic Diff–Weyl ghost one- and two-Ricci insertion traces and the generic physical fourth-order Hessian kernel, followed by integration and assembly of the five repository

parity-even third-curvature form-factor functions and coefficients, the parity-odd derivative carrier manifest, finite C^2 and absolute dressed $R(\widehat{g})^2$ normalization, global Paneitz Green data, and renormalized BV operator data fixing a complete Q_1 beyond its certified local Wess–Zumino, anomaly-induced, and flat-TT logarithmic contributions;

3. the induced residual operator $q_1 = \pi_{\mathrm{ext}} Q_1 \iota_{\mathrm{ext}}$;

4. nilpotency, adjacent-cohomology, Cartan, mixing, and pairing checks.

Until these land, residual transfer is forbidden. Same-background free-state and Hadamard constructions belong to a separate companion programme and are not claims of this paper. The classical-obstruction-to-interacting-BRST bridge also remains independently fail-closed.